

M5T1: Identifying Common Toxins — 11 min

Video File M5T1-Toxins.mp4 (toxin infographic; home audit map) · **Resources** Toxin Checklist.pdf (room-by-room audit, "list 3 to reduce" page) · **Xp** 90 total · Audit Home 40 · List 3 to Reduce 30 · Forum Audit Share 20

Target runtime 11 min · 1,264 spoken words · ≈ 8.4 min at 150 wpm · ≈ 9.4 min at a warm 135 wpm

0:00–0:30 **HOOK**

[VISUAL: Cynthia in her kitchen, holding a plastic food container in one hand and a glass one in the other.]

These two containers hold the same leftovers. One of them, warmed in the microwave, can release a chemical that acts like estrogen in the body. The other can't. That's the whole shape of this training: most of the toxins we care about have a simple swap sitting right next to them, and the hard part isn't the swap. It's seeing them in the first place. Toxins are sneaky. Let's make them visible.

0:30–5:30 **TYPES AND SOURCES**

[VISUAL: Toxin infographic: four categories with icons: plastics, pesticides, cleaning products, personal care; a fifth: air and water.]

Let me give you the main categories, where they live, and what the research says.

Category one: plasticizers. Bisphenol A, BPA, is used in hard plastics and in the linings of many food cans. Phthalates make plastics soft and carry fragrance in scented products. Both are

endocrine disruptors: they can mimic or block hormones. A study from the Harvard fertility cohort published in Human Reproduction found that women with higher urinary BPA had fewer eggs retrieved and a poorer response to IVF stimulation. In men, Russ Hauser's group at Harvard has repeatedly linked phthalate metabolites in urine to lower sperm quality, going back to a study in the journal Epidemiology in 2003. Where they hide: plastic food containers, especially heated; canned food linings; receipts printed on thermal paper, which transfer BPA to skin on handling; water bottles left in hot cars; plastic wrap on warm food; scented products.

[VISUAL: Infographic highlights pesticides.]

Category two: pesticides. In the same Harvard cohort, published in JAMA Internal Medicine in 2018, women who ate more high-pesticide-residue fruits and vegetables had a lower probability of pregnancy and live birth in fertility treatment than women who ate fewer, and the low-residue produce showed no such association. The message is not "eat fewer vegetables." It's "choose lower-residue ones," and Training Two will show you exactly how. Where they hide:

conventional strawberries, spinach, kale, apples, grapes; lawn and garden sprays; indoor pest sprays; the flea treatment on the dog.

[VISUAL: Infographic highlights cleaning products and air.]

Category three: cleaning products and indoor air. Cleaning sprays release volatile organic compounds; a large European study in the American Journal of Respiratory and Critical Care Medicine found that regular use of cleaning sprays was associated with a decline in lung function over twenty years comparable to smoking. And house dust turns out to be a reservoir: a review in Environmental Science and Technology by Mitro and colleagues found phthalates, flame retardants, and phenols in the dust of nearly every home tested. Where they hide: under the sink; in air fresheners, plug-ins, and scented candles; in dryer sheets; in dust on the floor where a crawling baby will one day be.

[VISUAL: Infographic highlights personal care.]

Category four: personal care. Parabens are preservatives in lotions, shampoos, and cosmetics; the word "fragrance" on a label can legally hide dozens of chemicals, including phthalates. Studies in Environmental Health Perspectives

have linked urinary paraben levels to sperm DNA damage in men and to markers of diminished ovarian reserve in women. Training Four is devoted to this cupboard.

[VISUAL: Infographic highlights the "big three": smoking, alcohol, and heat.]

And the ones that are so well known we forget them: tobacco smoke, including secondhand and vaping; alcohol, which we've addressed in Modules Two and Three; and for men, heat, from laptops on laps, hot tubs, and saunas, which I'll come back to in Training Five. The American College of Obstetricians and Gynecologists' committee opinion on environmental exposures lists these first, and so should we.

Let me put the science in proportion, because I don't want anyone leaving this training afraid of their own kitchen. These studies show associations, mostly in people who were already going through fertility treatment, and higher exposure is linked to somewhat worse outcomes, not to infertility on its own. Nobody's chances are decided by a food container. What the research does justify is exactly what we're doing: lowering the load where it's easy, for the months when it matters most, because the cost of the swaps is small and the potential

benefit reaches two bodies and possibly a third. Proportion, not panic, is the mood of this module.

5:30–9:30 **AUDIT DEMO**

[VISUAL: Home audit map: a floor plan; rooms light up in turn: kitchen, bathroom, laundry, bedroom, garage, entry.]

Now let's audit. The Toxin Checklist goes room by room, and I'd like you to do it with the checklist in one hand and nothing in the other, no bin bag yet. This is a looking week.

[DEMO: Cynthia opens her kitchen cupboards and drawers as she talks.]

Kitchen first, because it's where chemicals meet food and heat. Open the cupboard: how many plastic containers do you heat food in? Look at the pan: is the nonstick coating scratched? Look at the cans in the pantry: most linings contain BPA or a substitute. Look under the sink: how many spray bottles, and how many say "danger" or "warning"? Look at the counter: a scented candle? A plug-in freshener? Look at the receipts in the drawer. Tick what you find. Don't fix anything yet.

Bathroom: line up everything you put on your body in a day. Shampoo, conditioner, body wash, lotion, deodorant, sunscreen, makeup, perfume. Turn each one over. Circle

"fragrance" or "parfum," any word ending in "-paraben," and "phthalate" if it's listed, which it rarely is; fragrance is where it hides.

Laundry: detergent, fabric softener, dryer sheets, all usually fragranced. Bedroom: mattress age, plug-in fresheners, dust under the bed. Garage and garden: pesticides, herbicides, paint, solvents. Entry: do shoes come off at the door? Shoes track in pesticides and dust; a shoes-off home has measurably less of both.

[VISUAL: Slide: "Count, don't judge."]

By the end you'll have a count. I've done this audit with many people, and the number is usually higher than they expected, and the feeling is usually a mix of alarm and relief. Relief, because nearly every tick has a swap, and none of the swaps is expensive.

Here's a practical trick I use with the audit. Sort everything you found into three piles in your mind. Pile one: touches food or heat. Pile two: sits on skin or gets breathed all day. Pile three: everything else. Pile one is where the first swaps go, and it's usually the smallest pile: the containers, the pan,

the plastic wrap, the receipts. Most people can clear pile one in a single weekend for the cost of a set of glass containers. Pile two is Training Four. Pile three can take months, and that's fine.

9:30–10:30 PRIORITIZATION AND HEART-LED CLOSE

[VISUAL: Slide: "Prioritize: kitchen → personal care → cleaning → air and water."]

Prioritize by impact. First the kitchen, because heat and food contact are the most direct route into the body: stop heating food in plastic this week, and the rest can wait. Second, the products that sit on your skin all day: lotion, deodorant, the daily shampoo. Third, the cleaning products you breathe. Fourth, water and air, which take a little research and sometimes a little money. Your action is to list three things to reduce, not thirty. Three, from the top of that order. When those three are done, you'll do three more without being told to.

A word to anyone living in a shared home, a rental, or with family, where you don't control every product: audit what's yours and don't fight over the rest. Your own containers, your own toiletries, your own bedroom window. A shoes-off rule for your room if not the house. You'll lower your load

meaningfully with the things you control, and, in my experience, a calm example changes a household faster than an argument does.

This is for your future family. The home you're auditing today is the home a baby will crawl through, and every tick you turn into a swap is a little cleaner floor for them.

10:30–11:00 ACTION HANDOFF

[VISUAL: End card: "Toxin Checklist PDF → Audit your home (40 XP) → List 3 to reduce (30 XP) → Share your audit in the forum (20 XP)."]

Download the Toxin Checklist, audit your home room by room for forty XP, list your top three to reduce for thirty, and share your count and your three in the forum for twenty. Someone there has the same candle. Next, Training Two: clean eating, without the price tag.

ACTION CALLOUT

RESOURCE Toxin Checklist.pdf (room-by-room audit with tick boxes, the four categories explained, priority order, "my 3 to reduce" page). **ACTION 1:** Audit your home with the checklist (40 XP). **ACTION 2:** List three things to reduce (30 XP). **ACTION 3:** Share your audit count and your three in the forum (20 XP). **NEXT STEP:** Training 2 · Clean Eating Detox.